

Why the Minimal-Energy Velocity Is a Gradient

In Otto calculus, a velocity field v generates a tangent vector at a density ρ through the continuity equation:

$$\dot{\rho} = -\operatorname{div}(\rho v).$$

Different velocity fields can generate the same $\dot{\rho}$. Indeed, if w satisfies

$$\operatorname{div}(\rho w) = 0,$$

then

$$-\operatorname{div}(\rho(v + w)) = -\operatorname{div}(\rho v).$$

So v and $v + w$ represent the same tangent vector.

Choosing the Minimal-Energy Representative

The kinetic energy of a velocity field is

$$\int_{\mathbb{R}^d} |v|^2 d\rho.$$

Among all velocity fields producing the same tangent vector $\dot{\rho}$, Otto calculus chooses the one with minimal kinetic energy.

This is analogous to choosing the shortest vector among all representatives of an equivalence class.

Why It Is a Gradient

The gradient fields are orthogonal, in the weighted inner product

$$\langle u, v \rangle_\rho = \int_{\mathbb{R}^d} u \cdot v d\rho,$$

to the fields that do not move the density, namely fields w satisfying

$$\operatorname{div}(\rho w) = 0.$$

Indeed, for a smooth test potential ϕ ,

$$\int_{\mathbb{R}^d} \nabla \phi \cdot w d\rho = \int_{\mathbb{R}^d} \nabla \phi \cdot \rho w dx = - \int_{\mathbb{R}^d} \phi \operatorname{div}(\rho w) dx = 0.$$

Thus any velocity field can be decomposed formally as

$$v = \nabla \phi + w, \quad \operatorname{div}(\rho w) = 0.$$

The component w does not affect the tangent vector, but it increases kinetic energy:

$$\int |v|^2 d\rho = \int |\nabla \phi|^2 d\rho + \int |w|^2 d\rho.$$

Therefore the minimal-energy representative is obtained by setting $w = 0$:

$$v = \nabla \phi.$$

Interpretation

Only the compressive part of the velocity field changes the density. The weighted divergence-free part moves mass around without changing the density to first order, so it is extra motion. Wasserstein geometry discards that extra motion and keeps the gradient velocity.